MEDICAL INSURANCE COST PREDICTION USING MACHINE LEARNING

A Project Report Submitted in the partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

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Declaration

The Project Report entitled "**MEDICAL INSURANCE COST PREDICTION USING MACHINE LEARNING**" is a record of Bonafide work of **U. Jaya Sindhura – 2420030044, K. Ushasri – 2420030376, P. Smitha – 2420030583,** submitted in partial fulfillment for the award of B. Tech in Computer Science and Engineering to the K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

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This is to certify that the mini project based report entitled "**MEDICAL INSURANCE COST PREDICTION USING MACHINE LEARNING**" is a bonafide work done and submitted by **U. Jaya Sindhura – 2420030044, K. Ushasri – 2420030376, P. Smitha – 2420030583** in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in Department of Computer Science and Engineering, K L (Deemed to be University), during the academic year **2026-2027.**

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Abstract

Medical insurance plays a role in the way people manage healthcare expenses and the way individuals make financial decisions. Additionally, insurance information helps potential policyholders understand the factors that influence their costs. These factors help them estimate how much they may need to pay for medical insurance. Healthcare expenses have become an important concern, and if people are not currently considering these costs, it is time to understand what affects their insurance charges.

Most people consider several factors when estimating the cost of medical insurance. Many studies and datasets show that insurance charges are influenced by personal details and lifestyle factors such as age, sex, body mass index, number of children, smoking habits, and residential region. People tend to consider these details when trying to understand the amount they may have to pay for coverage. This is why machine learning methods are useful in medical insurance cost prediction. They make the estimation process easier by analyzing historical information and identifying patterns that may not be easy to notice manually. This approach applies to situations where we want to understand overall expenses and make decisions in a more time-efficient manner.

This is why the Medical Insurance Cost Prediction project uses machine learning to estimate insurance charges based on the information provided. The system focuses on factors such as age, BMI, smoking status, and other personal details to generate an estimated cost. Using the Medical Cost Personal Dataset helps the model learn from recorded insurance information. As you can see, predicting medical insurance costs involves factors that must be considered together. In fact, machine learning can make this estimation process more organized by analyzing data and identifying relationships between personal characteristics and insurance .

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Chapter 1
Introduction

1.1 Motivation

The main motivation of **"Medical Insurance Cost Prediction Using Machine Learning"** is to help individuals estimate their medical insurance expenses based on different personal and lifestyle factors. Medical insurance costs vary from person to person depending on age, gender, body mass index (BMI), number of children, smoking habits, and residential region. Since healthcare expenses are increasing, people need a better understanding of the factors that influence their insurance charges. This project uses machine learning to make the estimation process easier and helps users understand the possible cost of medical insurance based on the information they provide.

1.2 What is Machine Learning?

Machine Learning is a branch of Artificial Intelligence (AI) that enables computers to learn from data and make predictions or decisions without being explicitly programmed for every task. It uses algorithms to identify patterns and relationships in data and improves its performance through experience. Machine learning is widely used in different fields such as healthcare, banking, education, transportation, and business.

Machine learning algorithms can be used to solve different types of problems, including classification, regression, and clustering. In classification, the system predicts a category or class based on the input data. In regression, the system predicts a continuous numerical value. In clustering, the system groups similar data points based on their characteristics.

Among machine learning applications, supervised learning is commonly used when historical data contains both input features and known output values. The algorithm learns the relationship between the input and output data and uses this knowledge to predict results for

new data. In this project, machine learning is used to analyze medical insurance information and predict insurance charges based on different personal and lifestyle factors.

1.3 What is Medical Insurance Cost Prediction?

Medical Insurance Cost Prediction is the process of estimating the amount of money a person may need to pay for medical insurance based on different factors. These factors may include age, sex, body mass index (BMI), number of children, smoking habits, and residential region. Each factor can influence the insurance charges in different ways.

Medical insurance cost prediction uses historical insurance data to understand the relationship between personal details and insurance expenses. By analyzing this information, a machine learning model can identify patterns and estimate the expected insurance charges for a new individual.

Medical insurance cost prediction can be useful for individuals who want to understand possible insurance expenses and plan their healthcare budgets. It can also help demonstrate how data analysis and machine learning techniques can be applied to real-world financial and healthcare-related problems. The predicted amount is an estimate based on the dataset and model, not a confirmed insurance quotation.

1.4 What is a Dataset?

A **dataset** is a collection of related information organized in a structured format, usually in rows and columns. It is used to store, analyze, and process information for different purposes. In machine learning, datasets are important because they provide the information required for training and testing a model.

A dataset generally contains records and features. Each row represents an individual record, while each column represents a particular feature or attribute. The data can contain numerical values, categorical values, or both. Before using a dataset for machine learning, it may require preprocessing, such as handling missing values, converting categorical values into numerical form, and selecting relevant features.

In this project, the **Medical Cost Personal Dataset (insurance.csv)** is used for predicting medical insurance charges. It contains information such as age, sex, BMI, children, smoker, region, and charges. The first six attributes are used as input features, while charges is the target variable that the model aims to predict. The dataset helps the machine learning model learn the relationship between personal characteristics and medical insurance costs.

Chapter 2
System Requirements

2.1 Hardware Requirements

The following hardware is required to develop and run the project:
1. Computer or Laptop
2. Minimum 4 GB RAM
3. Processor: Intel i3 or above
4. Hard Disk: Minimum 500 GB
5. Internet connection

2.2 Software Requirements

The following software is used in this project:
1. Operating System: Windows 10/11
2. Programming Language: Python
3. Framework: Streamlit
4. Data Handling: Pandas
5. Data Visualization: Matplotlib
6. Dataset: CSV file
7. Web Browser: Google Chrome or Microsoft Edge

2.3 Development Tools

- Python IDLE or Visual Studio Code
- Streamlit
- Pandas library
- Matplotlib library

Chapter 3
Technology Stack

3.1 Python

Python is the main programming language used for developing the application. It is used to process user inputs, perform calculations and manage the application functions.

3.2 Streamlit

Streamlit is a Python framework used to create the web-based application. It provides an interactive dashboard through which users can access different features, such as disease search, treatment cost estimation, insurance details and claim history.

3.3 Pandas

Pandas is a Python library used to read and process medical cost data stored in a CSV file. It helps organize and manage the dataset.

3.4 Matplotlib

Matplotlib is a Python library used to create graphs and charts. It helps display medical cost information in a visual format.

3.5 Regular Expression (re)

The re module is a built-in Python module used to validate the phone number entered by the user during login.

3.6 CSV Dataset

The CSV dataset stores information such as disease names, treatment details and estimated medical costs. The application reads this data to display the required information.

3.7 Session State

Streamlit session state is used to maintain temporary information during application use. It stores details such as user information, selected insurance plans and claim records.

Chapter 4
System Architecture

The system architecture describes the structure and working of the Medical Insurance Cost Prediction application. It explains how the user interface, application logic and data storage work together to provide medical cost and insurance-related information.

4.1 Architecture Overview

The application is developed using Python and Streamlit. Streamlit provides the user interface, while Python handles user inputs and application operations. Medical treatment-cost information is stored in a CSV dataset. Session state maintains temporary user and claim information.

4.2 System Components

1. User
The user interacts with the application by entering login details and selecting the required features.

2. User Interface
The Streamlit dashboard provides different pages, such as Home, AI Disease Predictor, Treatment Cost Estimator, My Insurance Policy, Insurance Balance, Claim History and Disease Analytics.

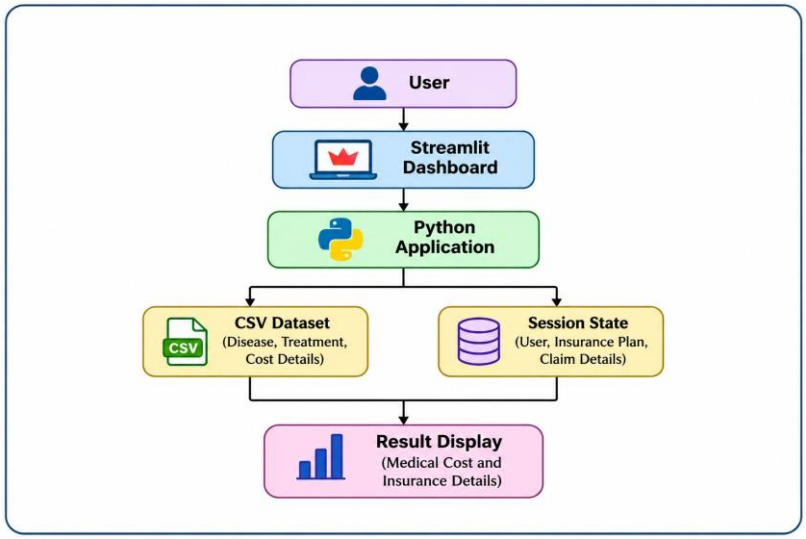
3. Application Logic
Python handles user inputs, performs disease keyword matching, estimates treatment costs and calculates the remaining insurance balance.

4. CSV Dataset
The CSV file stores disease names, treatment details and estimated medical costs. The application reads this information whenever required.

5. Session State
Streamlit session state maintains temporary information, including user details, selected insurance plans and claim records.

6. Result Display
The application displays treatment-cost information, insurance details, remaining balance, claim history and medical cost charts.

4.3 System Architecture Diagram



4.4 Working Process

1. The user opens the Medical Insurance application.
2. The user enters login details to access the dashboard.
3. The dashboard displays the available application features.
4. The user selects a required feature, such as disease search or treatment cost estimation.
5. Python processes the input and retrieves the required information from the CSV dataset or session state.
6. The application performs the required calculations.
7. The final information is displayed on the dashboard.

4.5 Data Flow

The data flow starts when the user enters information through the Streamlit dashboard. The Python application processes the input and accesses the required data. The processed results are then displayed to the user through the dashboard.

CHAPTER 5
DESIGN

5.1 User Interface Design

The application uses Streamlit to create an interactive web-based interface. The dashboard contains different pages that help users access the required features.

5.2 Home Page Design

The Home page acts as the main page of the application. It provides basic information about the project and allows users to navigate to different sections.

5.3 Login Page Design

The login section allows users to enter their name and phone number. The application checks whether the phone number contains 10 digits before allowing access.

5.4 Disease Predictor Design

The Disease Predictor section allows users to search for disease-related information. It uses keyword matching to identify disease names from the available dataset and display related information.

5.5 Treatment Cost Estimator Design

This section displays estimated medical treatment costs based on the selected disease or treatment information available in the CSV dataset.

5.6 Insurance Policy Design

The Insurance Policy section displays the available insurance plans and their coverage amounts. Users can select a plan according to their requirements.

5.7 Insurance Balance Design

This section displays the selected insurance plan and its remaining balance. The balance is calculated by subtracting approved claims from the plan coverage amount.

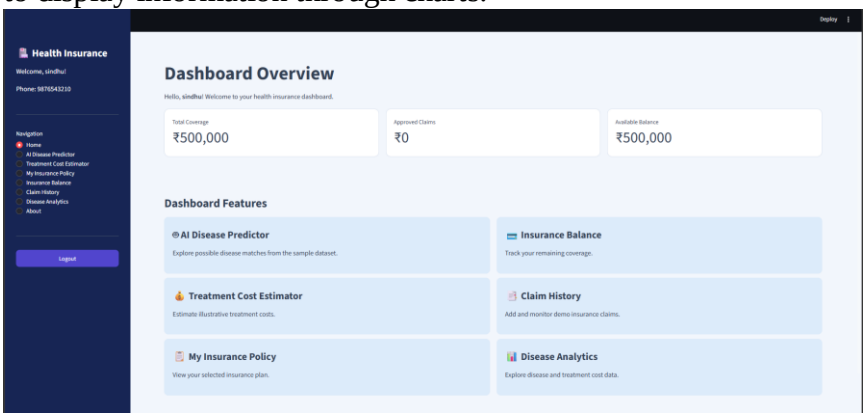
5.8 Claim History Design

The Claim History section displays claim details maintained during the application session. It helps users view their claim information.

CHAPTER 6
IMPLEMENTATION

6.1 Application Development

The application is developed using Python. Streamlit is used to create the web-based dashboard. Pandas is used to read and process medical cost information from a CSV file. Matplotlib is used to display information through charts.

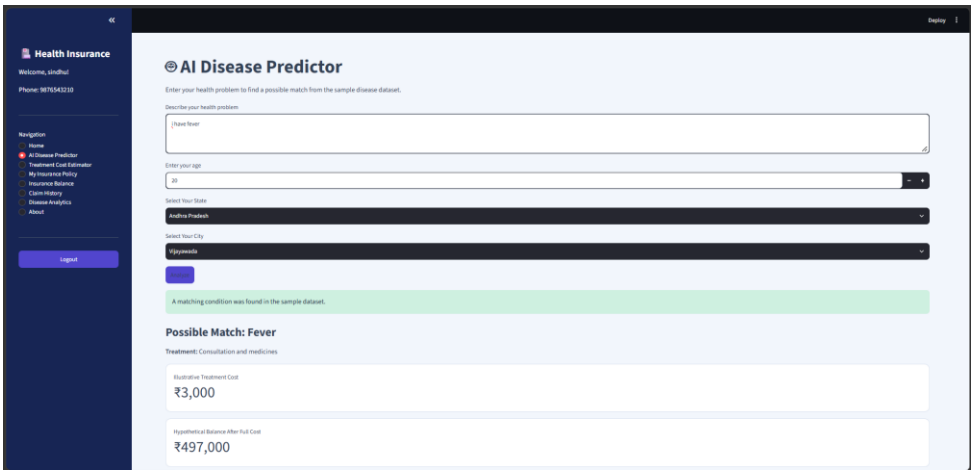


6.2 Login Implementation

The login section allows users to enter their name and phone number. The application checks whether the phone number contains 10 digits before allowing the user to access the dashboard.

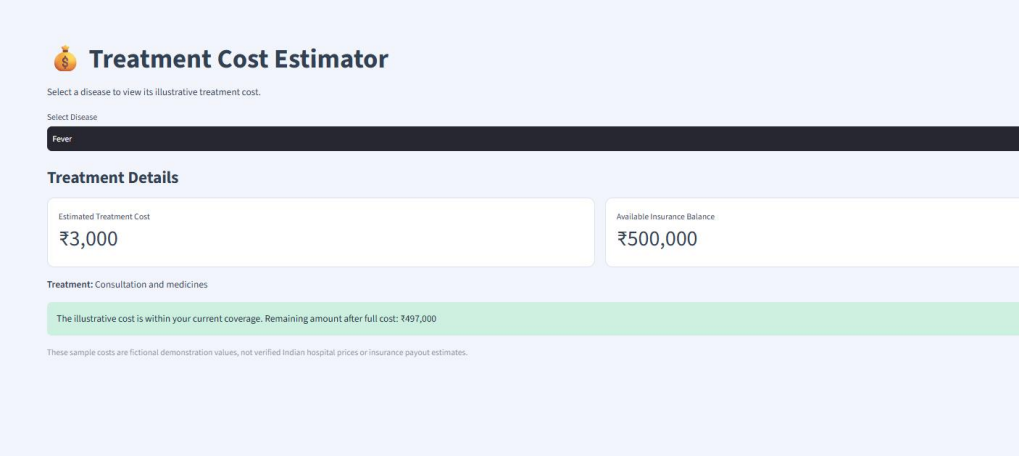
6.3 Disease Predictor Implementation

The Disease Predictor feature uses keyword matching to search for disease names in the available dataset. Based on the matching result, the application displays related disease and treatment information.



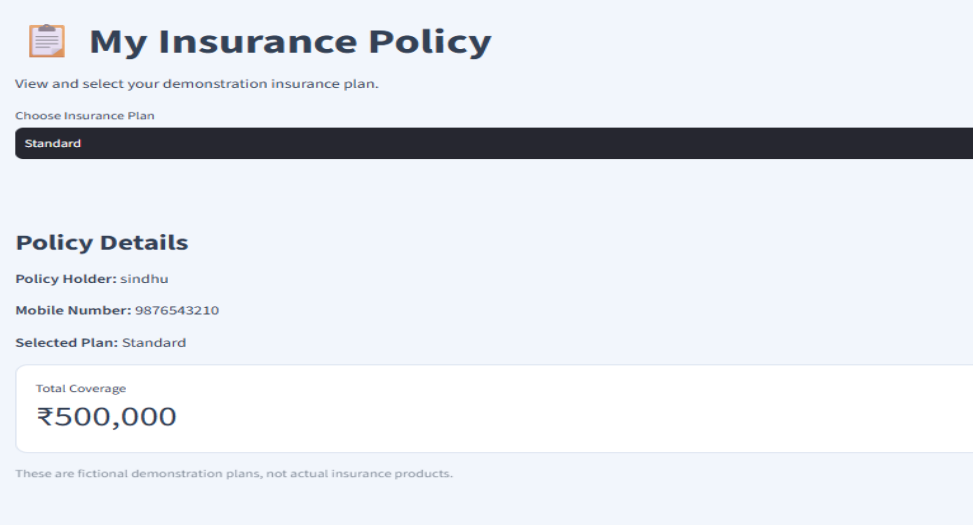
6.4 Treatment Cost Estimator Implementation

This feature displays estimated medical treatment costs based on the disease or treatment information available in the CSV dataset.



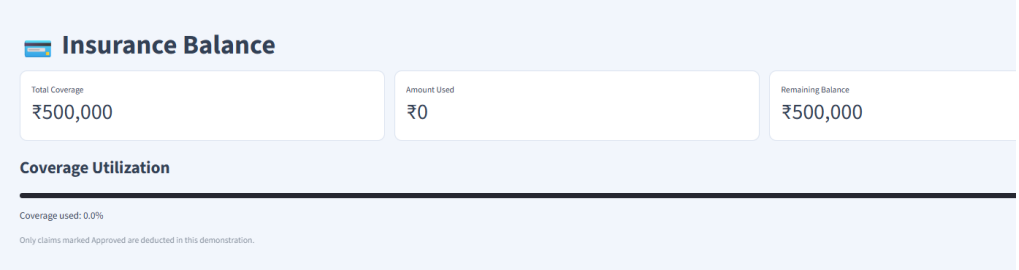
6.5 Insurance Policy Implementation

The application provides different insurance plans, such as Basic, Standard, Premium and Family. Each plan has a different coverage amount. Users can select a plan through the dashboard.



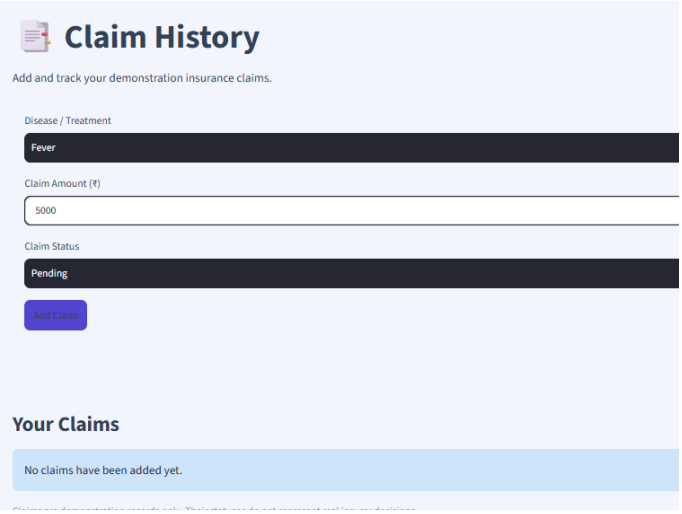
6.6 Insurance Balance Implementation

The Insurance Balance feature calculates the remaining coverage amount by subtracting approved claims from the selected insurance plan amount. It displays the available balance to the user.



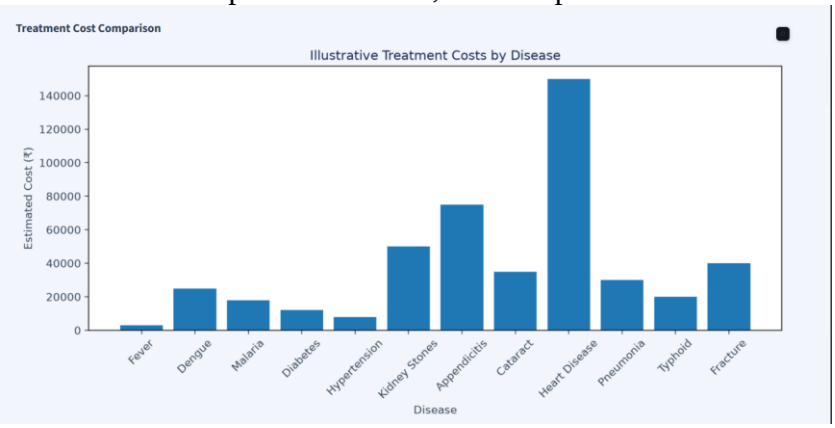
6.7 Claim History Implementation

The application maintains claim details using Streamlit session state. Users can view their claim information through the Claim History page.



6.8 Disease Analytics Implementation

The Disease Analytics feature displays medical cost information using tables and graphs. Pandas is used to process the data, and Matplotlib is used to create charts.



CHAPTER 7
FEATURES

The Medical Insurance application provides the following features to help users understand medical costs and insurance details.

7.1 User Login
Allows users to enter their name and phone number to access the application.

7.2 Home Dashboard
Provides easy access to all the main features of the application.

7.3 AI Disease Predictor
Uses keyword matching to search for disease names and display related information.

7.4 Treatment Cost Estimator
Displays estimated treatment costs using the available CSV dataset.

7.5 Insurance Management
Provides different insurance plans and displays the remaining coverage balance.

7.6 Disease Analytics and Claim History
Displays claim details and medical cost information through tables and charts.

CHAPTER 8
TESTING

Testing is an important process in software development. It is performed to check whether the Medical Insurance application works correctly. Different inputs are given to the application, and the displayed results are checked.

8.1 Login Testing

The login feature is tested by entering names and phone numbers. The application checks whether the phone number contains 10 digits.

8.2 Disease Predictor Testing

Different disease names are entered to check whether the application finds matching information from the available dataset.

8.3 Treatment Cost Testing

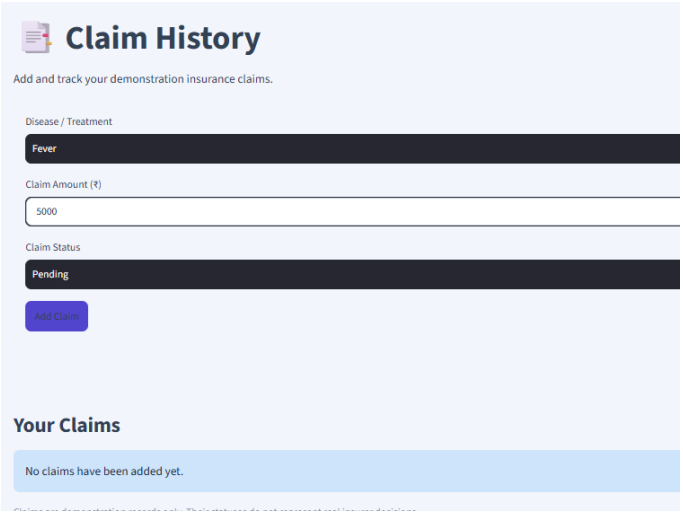
Different treatment details are selected to check whether the application displays the correct estimated costs from the CSV file.

8.4 Insurance Plan Testing

The available insurance plans are checked to verify whether their coverage amounts are displayed correctly.

8.5 Insurance Balance Testing

The balance calculation is checked to verify whether approved claim amounts are subtracted from the selected insurance coverage.



CHAPTER 9
DEPLOYMENT

Deployment is the process of running the Medical Insurance application so that users can access its features. The application is developed using Python and Streamlit.

9.1 Application Setup

Python and the required libraries are installed in the system. The application files and CSV dataset are kept in the project folder.

9.2 Running the Application

The application is started by entering the following command in the terminal:

```
bash
streamlit run app.py
```

9.3 Accessing the Application

After running the command, the application opens in a web browser. Users can enter their login details and access the dashboard.

9.4 Application Features

After deployment, users can access the Disease Predictor, Treatment Cost Estimator, Insurance Policy, Insurance Balance, Claim History and Disease Analytics sections.

CHAPTER 10
CHALLENGES AND LIMITATIONS

During the development of the Medical Insurance Cost Prediction application, some challenges and limitations were identified. These are explained below.

10.1 Challenges

1. Data Management: Organizing medical treatment and cost information in a CSV file was a challenge.

2. User Interface: Creating a simple and user-friendly dashboard required careful design.

3. Data Processing: Processing user inputs and displaying the correct information required proper handling.

4. Cost Estimation: Displaying suitable estimated treatment costs depended on the available dataset.

10.2 Limitations

1. The application uses a limited medical cost dataset.

2. The Disease Predictor uses keyword matching and is not a trained medical diagnosis model.

3. The treatment costs are estimates and may differ from actual hospital charges.

4. The insurance plans are demonstration plans and are not real insurance policies.

5. User and claim information is maintained temporarily using Streamlit session state.

6. The application requires further improvements for real-world use.

CHAPTER 11
FUTURE ENHANCEMENTS

The Medical Insurance Cost Prediction application can be improved in the future by adding new technologies and features. These improvements can make the application more useful, accurate and user-friendly.

Machine Learning Integration

A machine learning model can be added to predict medical insurance costs based on user details such as age, BMI, smoking habits and other factors. This can help provide more personalized cost predictions.

Improved Disease Predictor

The Disease Predictor can be improved by using advanced machine learning techniques instead of basic keyword matching. This can help provide better disease-related information.

Larger Medical Dataset

More disease names, treatment details and medical cost information can be added to the dataset. A larger dataset can help the application provide more information to users.

Database Integration

A database can be added to store user details, selected insurance plans and claim records permanently. This will help users access their information whenever they return to the application.

Online Deployment

The application can be hosted on an online server so that users can access it from anywhere through the internet without running it locally.

Improved User Interface

The dashboard can be improved with a better design, additional options and easy navigation. This can make the application more attractive and convenient to use.

Additional Insurance Features

More insurance plans and features can be included in the future. Users may also be provided with additional information about coverage, benefits and claim status.

CHAPTER 12
CONCLUSION

The Medical Insurance Cost Prediction project is developed to provide users with an easy way to understand medical treatment costs and basic insurance-related information. The main aim of this project is to create a simple and user-friendly web application where users can access different medical and insurance features in one place.

The application is developed using Python and Streamlit. Streamlit helps to create an interactive dashboard, while Python is used to handle user inputs and application operations. Pandas is used to read and process medical cost information from the CSV dataset. Matplotlib is used to display medical cost information through graphs and charts.

The application includes several useful features such as user login, AI Disease Predictor, Treatment Cost Estimator, My Insurance Policy, Insurance Balance, Claim History and Disease Analytics. These features allow users to search for disease-related information, view estimated treatment costs, select insurance plans and check their remaining coverage balance.

The Disease Predictor uses keyword matching to find disease names in the available dataset and display related information. The Treatment Cost Estimator provides estimated treatment costs based on the available data. The Insurance Policy section displays different plans, including Basic, Standard, Premium and Family plans. The Insurance Balance section calculates the remaining coverage amount based on approved claims. The Claim History section displays claim details maintained during the application session.

This project helped us gain practical knowledge of Python programming, Streamlit application development, data processing and data visualization. It also helped us understand how different software technologies can be combined to develop a useful web-based application.

During the development process, we learned how to organize data using CSV files, process user inputs, create interactive pages and display information in a clear format. We also understood the importance of testing the application to check whether its different features work properly.

Although the application provides basic medical cost and insurance information, it has some limitations. The disease search is based on keyword matching, and the treatment costs are estimates from the available dataset. The insurance plans are demonstration plans and do not represent actual insurance policies.

Overall, this project provides a simple platform for viewing medical treatment costs and basic insurance information. It helped us improve our technical knowledge and gain practical experience in developing a web-based application using Python and its supporting libraries.

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CHAPTER 14
Appendices

14.1 Importing Libraries

The following libraries are used in the project for dashboard development, data handling, and visualization.

```
import streamlit as st import pandas as pd
import matplotlib.pyplot as plt import re
```

14.2 Page Configuration

This code is used to configure the dashboard title, icon, and layout.

```
st.set_page_config(
page_title="Indian Health Insurance AI Dashboard", page_icon="·",
layout="wide"
)
```

14.3 Session State

Session state is used to maintain user login details, selected insurance plans, and claim information.

```
defaults = {
"logged_in": False, "user_name": "",
"user_phone": "",
"claims": [], "selected_plan": "Standard"
}

for key, value in defaults.items(): if key not in st.session_state:
st.session_state[key] = value
```

14.4 Insurance Plans

This code defines the different insurance plans and their coverage amounts.

```
plans = {
"Basic": 200000,
"Standard": 500000,
"Premium": 1000000,
"Family": 1500000
}
```

14.5 Dataset Loading

The Pandas library is used to load the treatment cost dataset from a CSV file.

```
@st.cache_data
def load_cost_data():
df = pd.read_csv("indian_medical_costs.csv") return df

cost_data = load_cost_data()
```

14.6 AI Disease Predictor

This code checks the user's entered health problem and searches for a matching disease in the sample dataset using keyword matching.

```
problem = st.text_area("Describe your health problem") if st.button("Analyze"):
matched = None
text = problem.lower()

for _, row in cost_data.iterrows(): disease = str(row["disease"]).lower()

if disease in text: matched = row break

if matched is not None: st.success("Matching condition found") st.write(matched["disease"])
```

14.7 Insurance Balance Calculation

This code calculates the approved claim amount and remaining insurance balance.

```
total_coverage = plans[ st.session_state.selected_plan
]

approved_claims = sum( claim["amount"]
for claim in st.session_state.claims if claim["status"] == "Approved"
)

remaining_balance = max( 0,
total_coverage - approved_claims
)
```

14.8 Claim History

This code is used to store and display the user's demonstration insurance claims.

```
st.session_state.claims.append({ "disease": disease, "amount": int(amount),
"status": status
})

claims_df = pd.DataFrame( st.session_state.claims
)

st.dataframe(claims_df)
```

14.9 Disease Analytics

Matplotlib is used to display a bar chart comparing estimated treatment costs for different diseases.

```
fig, ax = plt.subplots(figsize=(10, 5))

ax.bar(
cost_data["disease"], cost_data["estimated_cost"]
)

ax.set_xlabel("Disease")      ax.set_ylabel("Estimated      Cost      (₹)")
ax.set_title("Treatment Costs by Disease")

plt.xticks(rotation=45) plt.tight_layout()

st.pyplot(fig)
```